Tumour Proliferation and Receptor Status in TCGA-BRCA: Phase 1 Report

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Statistics and Its Applications, Phase 1 (Path B: TCGA-BRCA, Clinical + RNA-Seq)

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Abstract

This report documents the first phase of an individual project on breast cancer data from The Cancer Genome Atlas (TCGA-BRCA). We describe how 1,111 RNA-Seq files and the clinical tables were mapped to 1,095 patients, how the data were audited, and how an analytic cohort of 940 patients with complete ER, PR and HER2 status and age was defined. Tumour proliferation is measured by a 17-gene expression score. Descriptive comparisons show higher proliferation in ER- and PR-negative tumours, and patients excluded for incomplete data are not a random subset. The phase ends with a single statistical question for the later analysis.

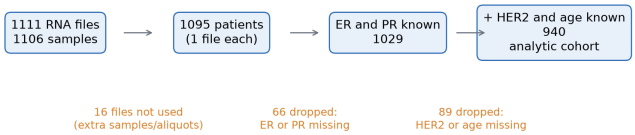


Figure 1. From downloaded files to the analytic cohort.

1. Data and design

We use TCGA-BRCA open-access data from the NCI Genomic Data Commons (release 46.0, downloaded 2026-10-21) [1, 2]. Two data types are combined: *clinical* tables (`clinical`, `follow_up`, `pathology_detail`, slide data) and *RNA-Seq* gene expression (STAR–Counts, GENCODE v36; analysed as $\log_2(\text{TPM}+1)$). The study is a retrospective observational analysis of existing data, without randomised intervention. TCGA patients are not a random sample, so results describe the patients studied rather than all breast cancer patients.

2. From files to patients

The files differ in what one row means: `clinical` is nested (5,546 rows for 1,098 patients: patient × diagnosis × treatment), `follow_up` is nested by follow-up and test, and each RNA file is one tissue sample. The analysis unit is therefore the *patient*. RNA files carry no patient ID; the GDC sample sheet links each file to a sample and patient. The 1,111 files belong to 1,106 samples and 1,095 patients (1,084 with one file, 6 with two, 5 with three). To avoid repeated observations of one patient, one file per patient is chosen by a fixed rule: smallest sample ID, then smallest aliquot ID, then file ID. The 16 unused files are kept in an audit table. All 1,095 patients also have clinical data; 3 patients have clinical data only and none has RNA only (Fig. 1).

3. Clinical audit

Tumour variables come from the one diagnosis flagged as primary disease (1,096 patients; two have none flagged). This removes a conflict in stage that otherwise affects 21 patients with several diagnosis records. Demographic fields were consistent within patients. Age is missing for one patient and stage for 26.

ER, PR and HER2 are stored as molecular-test rows in `follow_up`. ER and PR are taken from immunohistochemistry (IHC). HER2 is taken from definite FISH, otherwise from definite IHC; equivocal results and conflicting repeated tests are set to missing, and a conflicting FISH result is not replaced by IHC. ER and PR are available for 1,038 of 1,098 patients and HER2 for 958. HER2 is missing mainly because no test was recorded (114 patients), less often because the result was equivocal (24) or conflicting (2). These test rows have no date or timepoint, so they cannot be confirmed as primary-tumour results at diagnosis.

4. RNA audit and molecular variables

All 1,095 selected files contain the same 60,660 genes, TPM sums to 10^6 in every file, and library sizes range from 19.3 to 114.0 million counts. The share of unmapped or multimapping reads is unusually high in some samples (median 10.0%, maximum 60.8% in TCGA-C8-A12X-01A); 72 samples exceed the upper IQR fence of 15.2%. The available metadata do not explain this, and these samples were retained.

Three molecular variables are defined from a limited, prespecified gene set. The **proliferation score** is the mean $\log_2(\text{TPM}+1)$ of 17 genes (*CCNB1, CCNE1, CDC20, CEP55, EXO1, KIF2C, MELK, MYBL2, NUF2, PTTG1, RRM2, TYMS, UBE2C, UBE2T, BIRC5, ANLN, MKI67*). It shares 14 genes with the 18-gene Prosigna proliferation component [3, 4] but is project-defined, not a validated clinical signature. The 17 genes are co-expressed (median pairwise Spearman $\rho = 0.78$, range 0.54–0.93), which supports averaging them. The **luminal score** is the mean z-score of *ESR1, PGR, FOXA1, GATA3*, and **ERBB2** is $\log_2(\text{TPM}+1)$. *MKI67* mRNA is kept as a secondary measure; it is not the Ki-67 protein index.

5. Analytic cohort and descriptive results

Requiring ER, PR, HER2 and age leaves 940 of 1,095 patients (ER and PR known: 1,029; HER2 missing in a further 89). In this cohort 77.4% are ER-positive, 67.2% PR-positive and 19.4% HER2-positive; 63.7% are HR+/HER2−, 19.4% HER2-positive and 16.9% triple-negative. Median age is 59 years (IQR 49–68), 11 patients are men, and 57.6% have stage II disease.

The 155 excluded patients differ from the cohort in outcome-related variables but not in age or expression (Fig. 2): 33.1% vs. 10.7% are recorded as dead, 26.6% vs. 9.2% had tumour at last contact, and 32.2% vs. 24.1% have stage III–IV disease, whereas median age (58 vs. 59

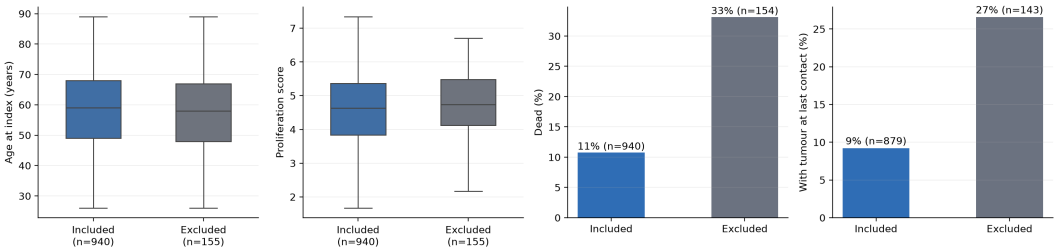


Figure 2. Patients with complete ER, PR, HER2 and age (included, $n = 940$) compared with the other RNA patients (excluded, $n = 155$).

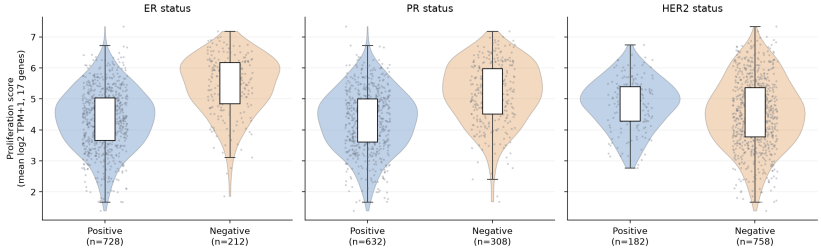


Figure 3. Proliferation score by ER, PR and HER2 status in the analytic cohort ($n = 940$). Boxes show median and IQR; points are patients.

years) and proliferation score (4.74 vs. 4.64) are similar. The complete-data cohort therefore does not represent all RNA patients.

The proliferation score is approximately symmetric (median 4.64, IQR 3.84–5.37, skewness -0.18). It is clearly higher in receptor-negative tumours (Fig. 3): median 5.57 in ER-negative vs. 4.38 in ER-positive (difference 1.19, Cohen's $d = 1.10$), and 5.28 vs. 4.33 for PR. The HER2 difference is small and opposite in direction (4.88 in HER2-positive vs. 4.54), and triple-negative tumours have the highest median (5.80, $n = 159$). These are descriptive findings only. Outliers by the $1.5\times$IQR rule were counted but kept, since extreme expression can be biological: 1 for the proliferation score, 5 for *MKI67*, 25 for the luminal score and 86 for *ERBB2*, whose high values are expected in HER2-positive tumours (median 9.0 vs. 6.8 in HER2-negative).

6. Final question and limitations

The data support the following question: *among TCGA-BRCA breast cancer patients, does adding RNA-derived molecular information (a luminal-program score and ERBB2 expression) to clinical information (ER, PR and HER2 status and age) explain additional variation in tumour proliferation, measured as a 17-gene expression score?* The multi-gene score was preferred to *MKI67* alone after exploratory plots, because a single gene is a noisier measure.

The main limitations are: (i) the 940-patient cohort is a non-random subset with better recorded outcomes; (ii) receptor tests are undated and HER2 combines FISH and IHC; (iii) bulk RNA mixes tumour and non-tumour cells, and some samples have a high share of unmapped reads; (iv) the 17-gene score is project-defined.

References

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[3] Parker JS *et al.* Supervised risk predictor of breast cancer based on intrinsic subtypes. *J Clin Oncol*. 2009;27:1160–1167.

[4] Buus R *et al.* Development and validation for research assessment of Oncotype DX Breast Recurrence Score, EndoPredict and Prosigna. *npj Breast Cancer*. 2021;7:15.